SwiftLog Logistics Supply Chain Analytics and Business Intelligence

Executive Business Intelligence Report | Prepared by Olabisi Amusan | AnalystLab Africa Consulting, Week 5

1. Executive summary

SwiftLog processed 5,000 orders between 1 January and 8 September 2026 and earned ₦14.44M in revenue from non-cancelled orders. Demand is healthy: customers are loyal (96% order more than once) and revenue is well spread across customers and products. The main problem is operational. Only 12.22% of shipments arrive with no delay, one in three orders is cancelled and one in three is still pending. Returns run at 9.46% of orders, and inventory turns over very slowly (0.08). Our twelve recommendations focus on delivery reliability, order completion, stock control and data quality.

2. Business context and objectives

SwiftLog operates 15 warehouses (Lagos 8, Abuja 4, Kano 3) and serves customers in Ibadan, Kano, Lagos and Abuja. Management asked for a solution to monitor performance, improve delivery, reduce costs, improve inventory and warehouse operations, evaluate suppliers, reduce returns and support executive decisions.

3. KPI summary

KPI	Value	KPI	Value
Total revenue	₦14,439,218	On-time delivery rate	12.22%
Total orders	5,000	Average delivery delay	3.57 days
Total customers	991	Return rate (orders)	9.46%
Average order value	₦4.28K	Inventory stock level	1,006,935 units
Repeat customer rate	95.96%	Inventory turnover	0.08

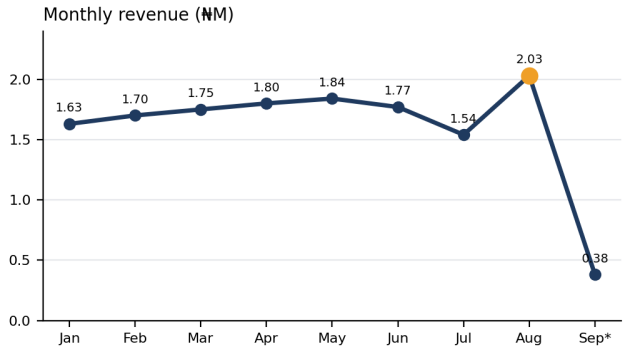
4. Data and preparation

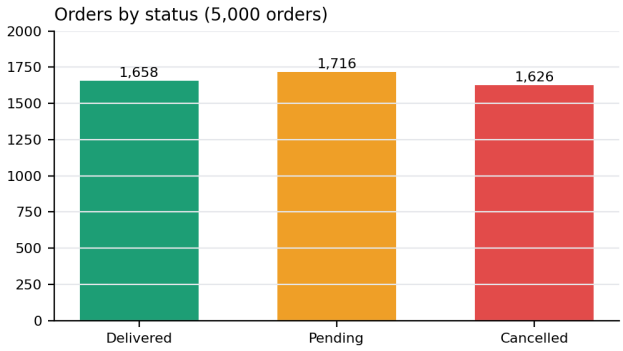
Eleven CSV files were loaded into Power BI and cleaned in Power Query: data types set, dates parsed, text trimmed, Month, Quarter and Year columns added, 3 duplicate inventory rows removed and repeated warehouse-product stock rows summed. A star-style model links Customers, Orders, Order_Items, Products, Shipments, Delivery_Performance, Warehouses, Inventory, Employees and Returns. Revenue is Quantity × UnitPrice and excludes cancelled orders. A shipment is On Time only when DelayDays is 0, because the source Status column contradicts DelayDays on 1,203 shipments. More than 25 DAX measures drive the five report pages (Executive, Customers, Inventory, Delivery, Suppliers).

5. Key findings

Demand and customers.

- Revenue is led by Fashion (₦5.41M, 37.5%), then Food (₦4.74M) and Electronics (₦4.29M).
- August is the peak month: 654 orders (463 completed) and ₦2.03M revenue. September is low only because the data stops on 8 September.
- Revenue is not concentrated: the top 10 products make up 8.4% of revenue and the top 10 customers 3.3% (the top 10% of customers earn about 23%).
- Customers are loyal: 95.96% placed more than one order, 56% placed five or more, and the average is 5.05 orders per customer. Kano earns the most revenue (₦3.87M).

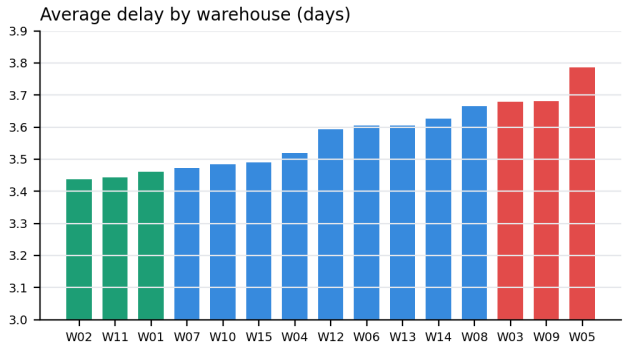


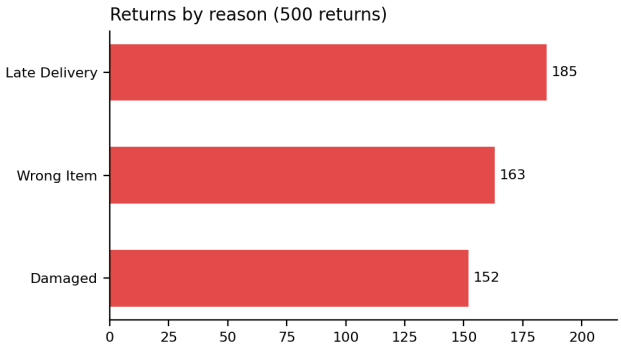


* September covers 1 to 8 September only.

Order flow. Of 5,000 orders, 1,658 are delivered, 1,716 are pending (₦7.33M) and 1,626 are cancelled (₦7.23M of order value). Only ₦7.11M of the reported revenue comes from delivered orders. In addition, 161 returns are linked to cancelled orders, which should not be possible.

Delivery. Only 611 of 5,000 shipments (12.22%) arrive with no delay, and 39% are five or more days late. Delay is network-wide: every city averages 3.47 to 3.64 days (Kano highest, Lagos lowest), and delays are spread evenly across 1 to 7 days. The best warehouses are W02, W11 and W01 (W01 has the best on-time rate at 14.2%). The weakest are W05 (3.79 days), W09 and W03 (W03 on-time rate 9.9%). The gap between best and worst is about 0.35 days.





Returns. There were 500 returns on 473 orders. The reasons are Late Delivery (185, 37%), Wrong Item (163, 33%) and Damaged (152, 30%). Return rates are almost identical across categories (Food 9.65%, Electronics 9.60%, Fashion 9.33%), so this is a company-wide issue. Product-level return rates are estimates because the Returns data has no ProductID (the most returned products are P231, P130 and P040).

Inventory and warehouses. SwiftLog holds 1,006,935 units, yet 264 product-warehouse lines hold fewer than 100 units (142 below 50, three at zero) while inventory turnover is only 0.08. W12 holds the most stock (77,517 units) and W01 the least (56,539). Workload also varies: W14 and W12 carry about 7,750 units per employee, against about 3,770 at W01.

Suppliers. Supplier performance could not be scored. Products and Shipments carry no SupplierID, so no supplier can be linked to a delay or a product. The 150 suppliers are listed on the dashboard and the gap is addressed in the recommendations.

6. Recommendations

#	Area	Finding and action
1	**Delivery optimisation**	On-time is 12.22%. Set a delivery SLA, review carriers and routes, report on-time rate weekly (target above 40% in six months).
2	**Warehouse efficiency**	W05, W09 and W03 are slowest. Run a process review and copy the practices of W02, W11 and W01.
3	**Cancellations**	32.5% of orders (₦7.23M) are cancelled. Record a reason for every cancellation (target below 20%).
4	**Pending backlog**	1,716 orders (₦7.33M) are pending. Add an ageing report and an escalation rule.
5	**Inventory management**	264 lines hold under 100 units. Set reorder points and replenish these first, starting with the 142 under 50.
6	**Stock balance**	Turnover is 0.08. Promote slow stock and rebalance between warehouses (W12 77,517 units vs W01 56,539).
7	**Operational cost**	Rebalance staff towards W14 and W12, and cut repeat deliveries by fixing late delivery and returns.
8	**Return reduction**	Late delivery is the top reason (37%). Fix delivery first, then add a pick-and-pack check (wrong item) and better packaging (damage).
9	**Return data quality**	Add ProductID to Returns and block returns on cancelled orders (161 exist today).
10	**Supplier improvement**	Add SupplierID to Products and Shipments, then build a supplier scorecard on delay and return rates.
11	**Customer retention**	Reward top customers (C0223, C0773, C0807) and move more customers into the five-or-more-orders group (56% today).
12	**Growth and monitoring**	Grow Fashion (37.5%) and Kano (₦3.87M), prepare stock and staff before the August peak, and review this dashboard weekly with KPI alerts.

Expected impact (illustrative). Better on-time delivery should reduce late-delivery returns. Fewer cancellations and a shorter backlog move more of the ₦14.44M towards delivered (recovering 25% of cancelled value would add about ₦1.8M). Reorder points reduce stock-outs, and rebalancing stock and staff reduces holding and handling cost. Targets are proposals for management, not forecasts.

7. Limitations

- Supplier performance could not be scored: the data has no link between suppliers and products or shipments.
- Warehouse utilization uses an assumed capacity (a What-If slider from 80,000 to 150,000 units per warehouse) because capacity is not in the data.
- Average delivery time could not be calculated because there are no shipment or delivery dates, so delay days is used.
- Product-level return rates are estimates: a return is credited to every product in the returned order.
- Delivery figures include shipments linked to cancelled orders, as in the source data. Inventory turnover counts units on all orders, including cancelled.
- September is a partial month (to 8 September).

8. Conclusion

SwiftLog has a loyal customer base and healthy demand. The biggest gains lie in delivery reliability, order completion and stock management, and closing the four data gaps (supplier link, shipment dates, warehouse capacity and return product) will make the next round of analysis much more precise.